

**SIMATS ENGINEERING
ASSESSMENT TOOL 1**

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COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)

Assessment Tool Weightage: 40%

Description about the assessment tool:

Short-answer questions effectively measure conceptual understanding, logical reasoning, and the ability to apply theoretical knowledge without requiring lengthy descriptive responses. This assessment method is also efficient for evaluating multiple OSI-related concepts within a limited time while ensuring objective and consistent evaluation.

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every

window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10. A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Q ₁ L ₃ L ₂ L ₁	Q ₂ L ₃ L ₂ L ₁	Q ₃ L ₃ L ₂ L ₁	Q ₄ L ₃ L ₂ L ₁	Q ₅ L ₃ L ₂ L ₁	Q ₆ L ₃ L ₂ L ₁	Q ₇ L ₃ L ₂ L ₁	Q ₈ L ₃ L ₂ L ₁	Q ₉ L ₃ L ₂ L ₁	Q ₁₀ L ₃ L ₂ L ₁
2.25	2	2	2.25	2	2.25	2.25	2	2	2.5

Understanding
45%

Explanation
35%

Clarity
20%

1	1.75	1	0.5	1.75	1	1	1.75	1	1
0.5	1	1	1	0.5	1	1	1	1	1
3.75	4.75	4	3.75	4.25	4.25	4.25	4.75	4	4.5

$$\frac{42.25}{50}$$

$$84.5$$

SA02
A Given
fil.

1. A given

file size = 150MB

Bandwidth = 50mbps

Segment size = 1500 byte

data link layer = 40 bytes

$$\begin{aligned}
 * \text{Total file size} &= \text{file size} \times 1,000,000 \\
 &= 150 \times 1,000,000 \\
 &= 150,000,000
 \end{aligned}$$

$$\begin{aligned}
 * \text{Number of segments} &= \frac{\text{Total file size}}{\text{Segment size}} \\
 &= 150,000,000 \div 1500 \text{ (or)} \frac{150,000,000}{1500} \\
 &= 100,000 \text{ segments}
 \end{aligned}$$

* one segment can form one frame, so
100,000 frames

$$\begin{aligned}
 * \text{total data sent} &= \text{file size} + \text{over head total} \\
 &= 150 \text{MB} + 4 \text{MB} \\
 &= 154 \text{MB}
 \end{aligned}$$

$$* \text{total over head} = 100,000 \times 40 = 4,000,000 \text{ bytes (4MB)}$$

$$\text{Convert to bits} = 154 \times 8 = 1232 \text{ Mb}$$

$$* \text{Transmission time} = \frac{1232 \text{ Total data sent in bits}}{\text{Band width}}$$

$$= \frac{1232}{50}$$

$$= 24.64 \text{ sec}$$

2A given

Window size = 6 frames

Total frames = 48

frame size = 1024 bytes

$$\text{Number of windows} = \frac{\text{total frames}}{\text{window size}}$$

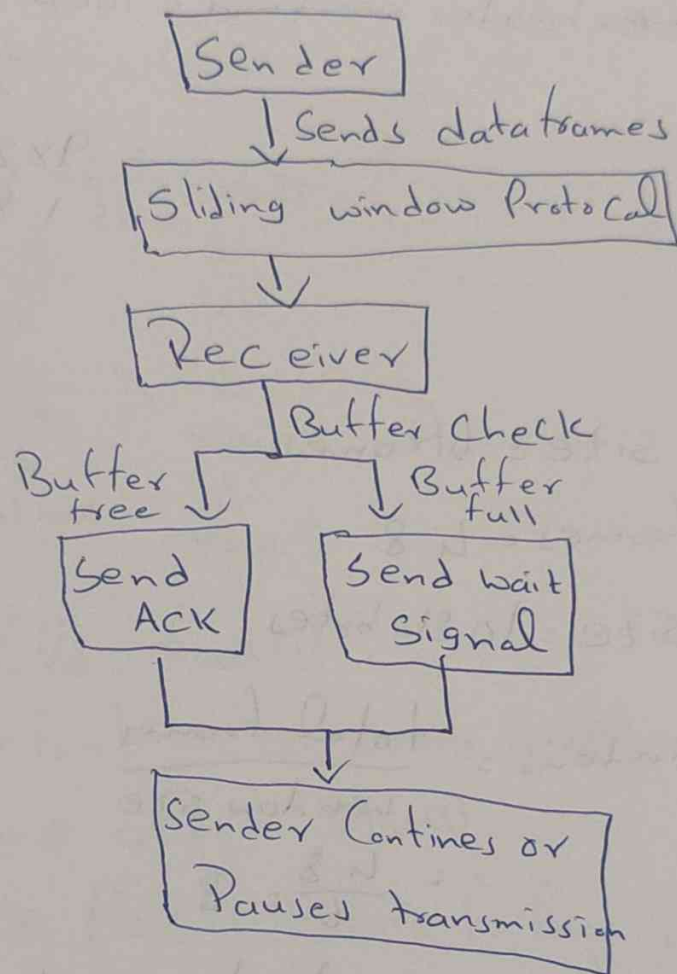
$$= \frac{48}{6} = 8 \text{ windows}$$

one frame is lost in every window

total retransmissions = 8 windows

*flow Control ensures that the Sender transmits data only at a rate the receiver can handle. This avoids buffer overflow, minimizes Packet loss, and reduces network traffic caused by retransmissions. Therefore, data transfer becomes smooth, efficient and accurate

flow chart:



3A given

IP Packet size = 12,000 bytes

MTU = 1500 bytes

IP header = 20 bytes

Max data in each = MTU - IP header

$$= 1500 - 20$$

$$= 1480 \text{ bytes}$$

Number of fragments = $\frac{\text{IP Packet size}}{\text{Max data in each}}$

$$= \frac{12,000}{1480} = 9 \text{ fragments}$$

$$\begin{aligned} \text{Total } \cancel{\text{overheader}} \text{ overhead} &= \text{No. of fragments} \times \text{IP header} \\ &= 9 \times 20 \\ &= 180 \text{ bytes} \end{aligned}$$

Q. A given

Window size = 6 frames

total frames = 48

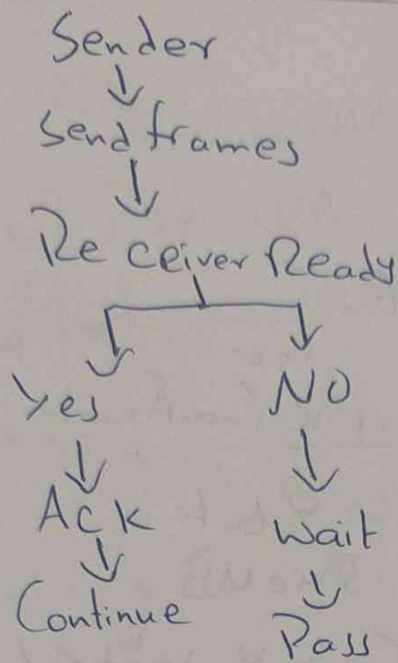
frame size = 1024 bytes

$$\begin{aligned} \text{No. of windows} &= \frac{\text{total frames}}{\text{window size}} \\ &= \frac{48}{6} = 8 \end{aligned}$$

One frame is lost in each window

$$\begin{aligned} \text{No. of windows} &= \text{Total retransmissions} \\ &= 8 \end{aligned}$$

Flow Control: Flow Control balances the speed of the sender and receiver. It prevents data loss by allowing transmission only when the receiver is ready.



5A Given

Video file = 600MB

Check Point after every 60MB

Failure after 390MB

* Check Points before failure = (60, 120, 180, 240, 300, 360) = 6

Last Check Point = 360

data retransmit = $390 - 360 = 30\text{MB}$

Session layer: the Session layer creates Check Points during Communications. If a failure occurs, data transfer resumes from the last Check Point instead of starting over

6.A Given

Original file = 900MB

Compression = 60%

Encryption overhead = 5%

$$\begin{aligned}\text{Compressed size} &= \frac{100 - \text{Compression}}{100} \times \text{Original size} \\ &= 900 \times 60\% \\ &= 540\text{MB}\end{aligned}$$

$$\begin{aligned}\text{Encrypted size} &= \text{Compressed size} \times 100\% + 5\% \\ &= 540 \times 1.05 \\ &= 567\text{MB}\end{aligned}$$

$$\begin{aligned}\text{Percent reduction} &= \left(\frac{900 - 567}{900} \right) \times 100 \\ &= 37\%\end{aligned}$$

Presentation layer:

Original data

↓
Compress

↓
encrypt

↓
Send data

→ The Presentation layer Compress data to reduce its size and encrypt it to keep the information secure during transmission

Given

data = 3GB (or) 3000MB

Request size = 3MB

Through Put = 120Mbps

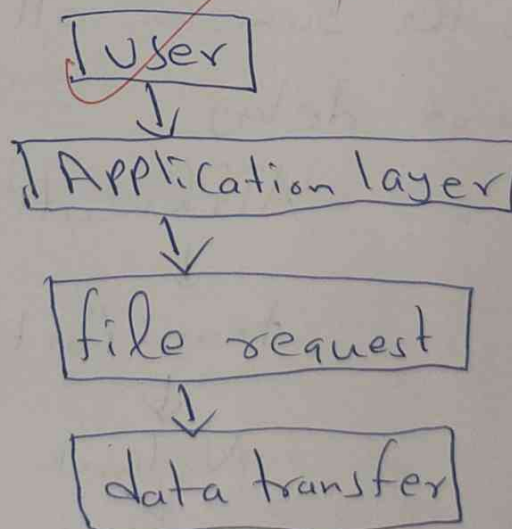
$$\text{No. of req} = \frac{\text{data}}{\text{Request size}} = \frac{3000}{3} = 1000$$

$$\begin{aligned} \text{data in bits} &= \text{data} \times 8 = 3000 \times 8 \\ &= 24000 \text{ Mb} \end{aligned}$$

$$\text{Transmission time} = \frac{24000}{120}$$

$$= 200 \text{ sec}$$

Application layer: the application layer provides services like file transfer and communication btw users and the network



8A given

Application = 4ms

transport = 3ms

Network = 5ms

data link = 2ms

Physical = 1ms

Progration delay = 18ms

transmission delay = 12ms

Total delay = $4 + 3 + 5 + 2 + 1 + 18 + 12$

Total end to end delay = 45ms

End to end delay: - each OSI layer adds a small Processing delay. The total Communication delay is the sum of all Processing and transmission delay

Application



trans Port



Network



data link



Physical

total delay

given

Useful data = 80MB

frame size = 1600 bytes

Overhead = 80 byte/frame

Useful data Per frame = frame size - useful data
 $= 1600 - 80 = 1520$ bytes

total useful data = 80,000,000 bytes

frames required = $\frac{80,000,000}{1520}$
 $= 52,632$ frames

total overhead = frames req $\times$ overhead = $52,632 \times 80$
 $= 4,210,560$ bytes

efficiency = $\frac{1520}{1600} \times 100 = 95\%$

Proto Col overhead: Proto Col overhead adds Control information to every frame. Although it supports reliable Communication, it slightly reduces transmission efficiency.

Useful data

↓
Add header

↓
transmit frame

↓
Received data

10A given

Original file = 240 MB

Compression = 30%

Segment size = 1400 bytes

Overhead = 60 bytes/frame

Compressed file size = o.f x remaining compression
 $= 240 \times 70$
 $= 168 \text{ MB}$

Compressed size = 168,000,000 bytes

Segments = $\frac{168,000,000}{1400} = 120,000$

Protocol overhead = $120,000 \times 60$
 $= 7,200,000 \text{ bytes (or) } 7.2 \text{ MB}$

final transmitted data = $168 + 7.2$
 $= 175.2 \text{ MB}$

Multiple OSI layers; different OSI layers work together to Compress, divide, frame and transmit data securely and reliably from Sender to receiver

